

Capítulo 4

Chapter Precap

- Discutir el análisis de datos compuestos por múltiples observaciones de los miembros de “un grupo”.
- Introducir y explicar los siguientes conceptos:
 - Modelos 'multinivel’.
 - Datos de "medidas repetidas".
 - Sin agrupación, agrupación completa y agrupación parcial adaptable.
 - Efectos 'aleatorios' y 'fijos’.
- Ajustamos un modelo multinivel con una estructura más apropiada para nuestros datos de medidas repetidas que los modelos del capítulo anterior.
- Simulamos datos de medidas repetidas en función de los parámetros estimados por nuestro modelo y vemos cómo la exclusión de diferentes componentes afecta a nuestros datos simulados.

Repeated Measures Data

- Observaciones múltiples de una "fuente" o "unidad experimental" determinada.
- Las observaciones no son independientes, lo que causa un problema si actuamos como si lo fueran.
- Por ejemplo, nuestro experimento contó con 139 observaciones de 15 oyentes diferentes (y 15 observaciones para cada hablante).

Last Chapter: One Big Pile of Data

$$\text{height}[i] \sim \mathcal{N}(\mu[i], \sigma)$$

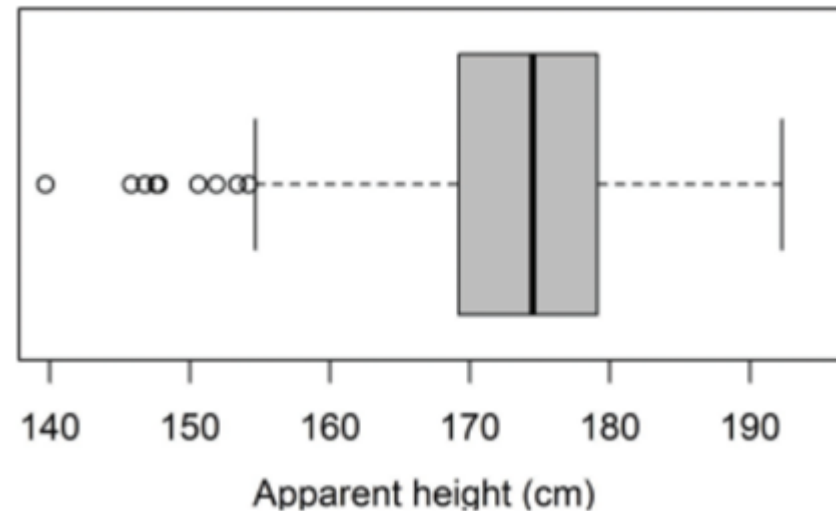
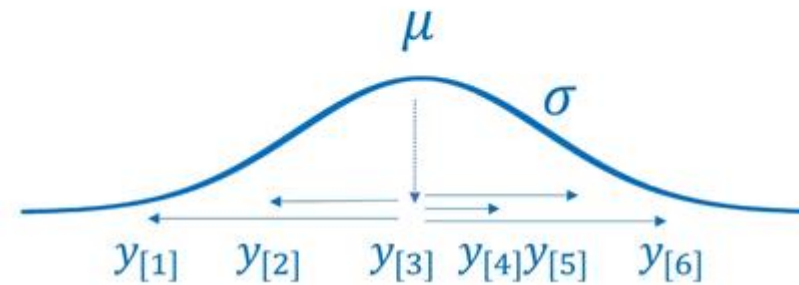
$$\mu[i] = \text{Intercept}$$

Priors:

$$\text{Intercept} \sim \mathcal{N}(176, 15)$$

$$\sigma \sim \mathcal{N}(0, 15)$$

“Unilevel” Model



Repeated Measures Data

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$

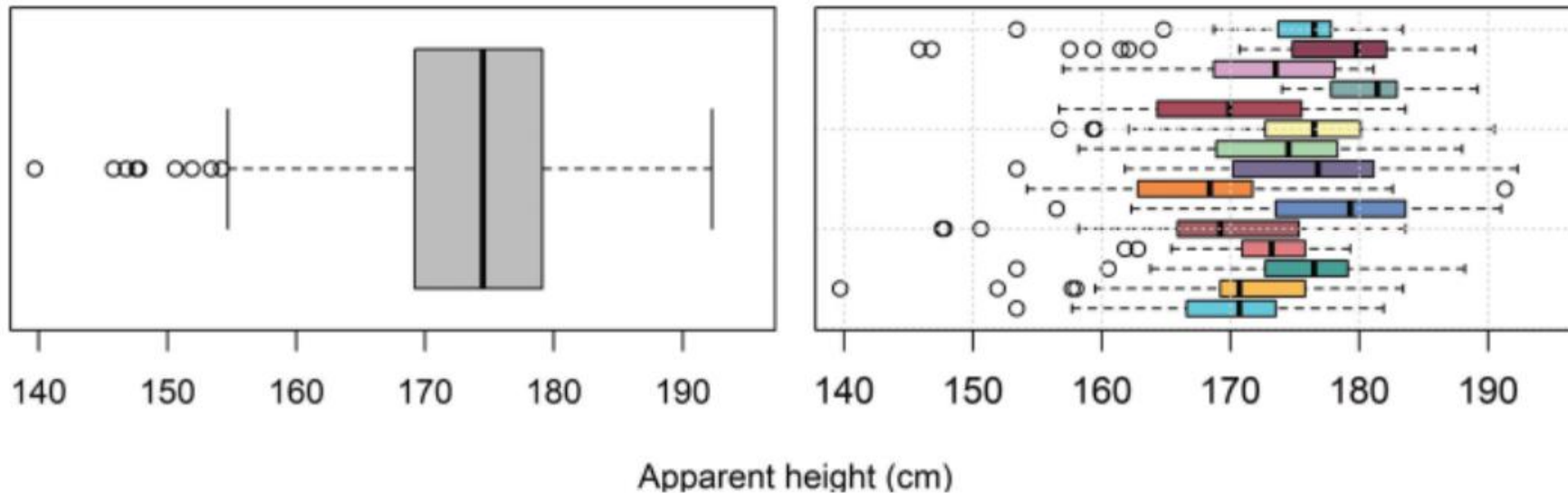
$$\mu_{[i]} = \text{Intercept}$$

Priors:

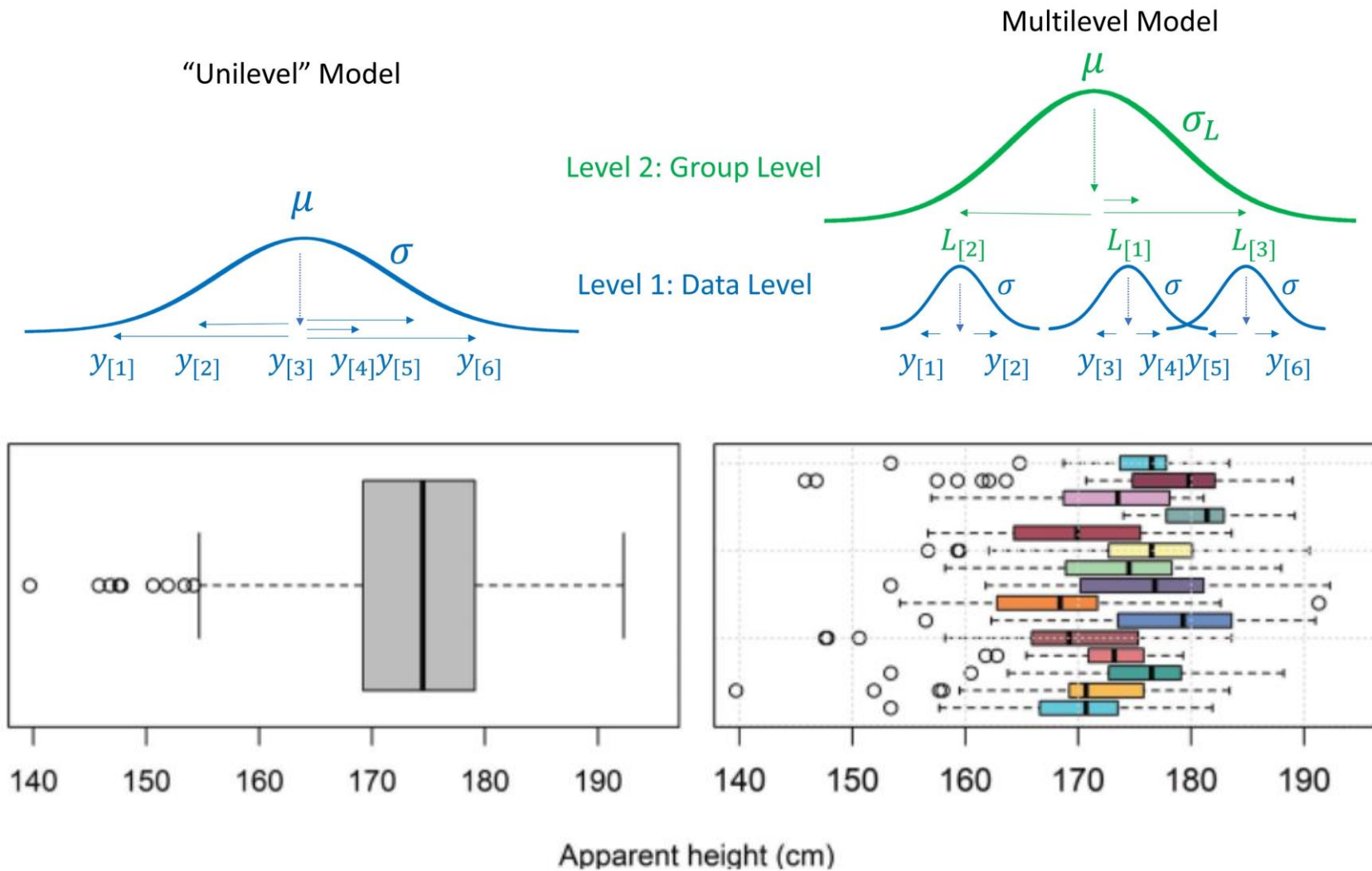
$$\text{Intercept} \sim N(176, 15)$$

$$\sigma \sim N(0, 15)$$

?

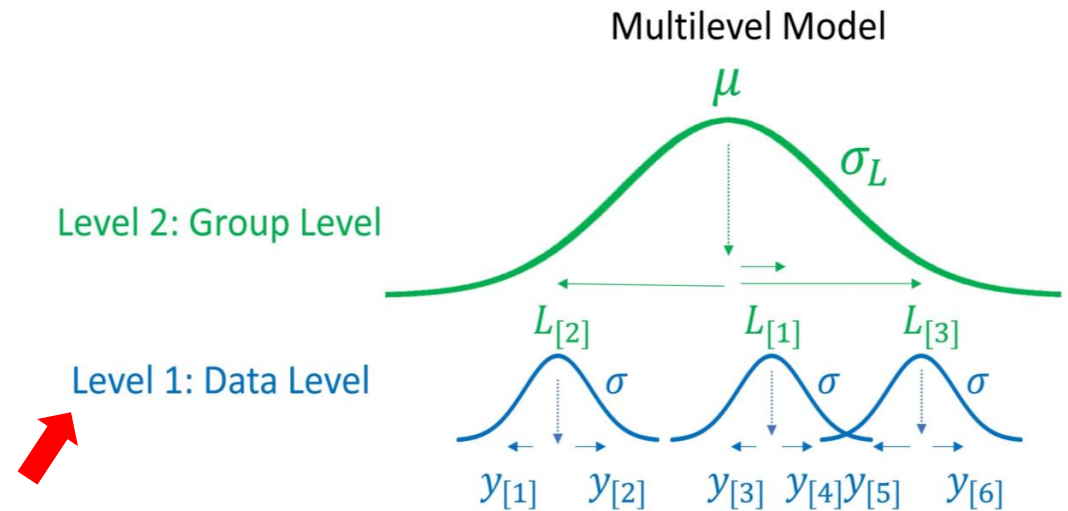


Repeated Measures Data: Levels of Variation



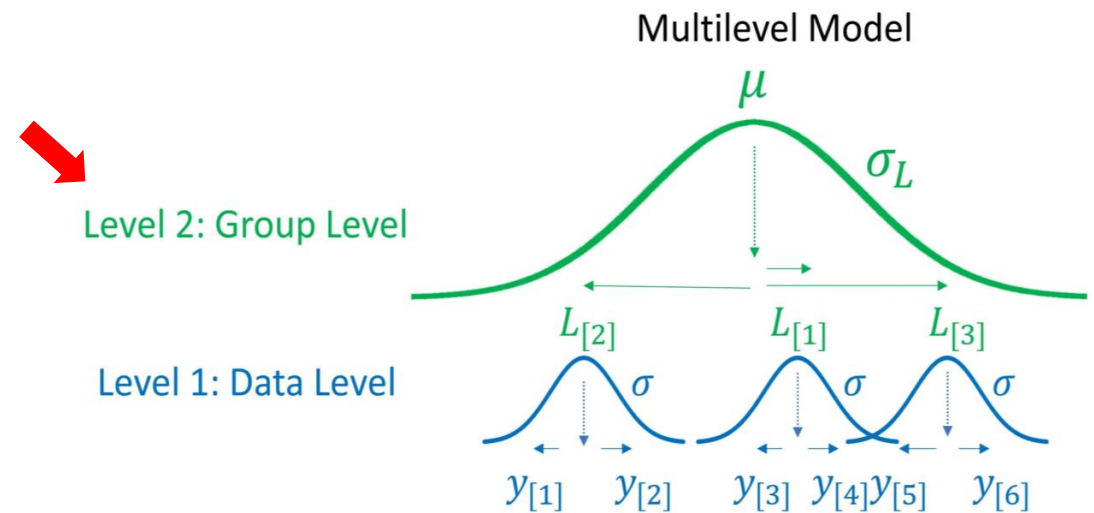
‘Levels’ of Variation: The ‘Lower’ level

- La distribución “data level” *dentro* de un clúster/unidad/sujeto determinado.
- La distribución condicional de datos dada una fuente específica de datos.
- Ejemplo: Una sola persona tendrá una distribución de tiempos de reacción para una tarea experimental.



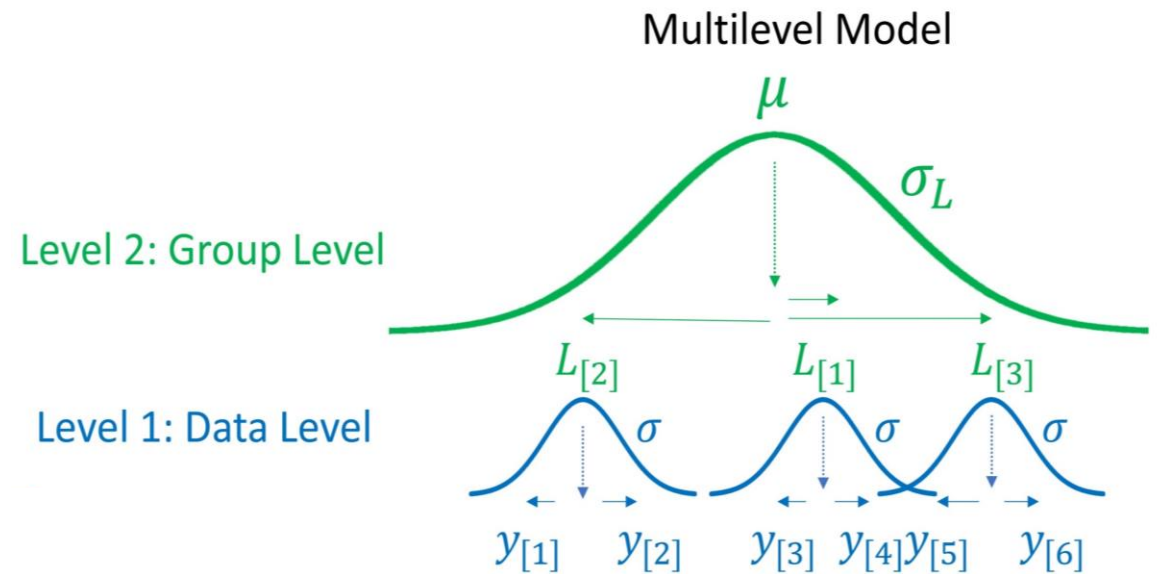
‘Levels’ of Variation: The ‘Upper’ level

- La distribución de parámetros “group level” en diferentes clústeres/unidades/sujetos.
- Distribución marginal de los parámetros entre todas las fuentes de datos.
- Ejemplo: Habrá una distribución de los tiempos de reacción promedio para una tarea experimental, entre los participantes.



Multilevel Models

- Los modelos multinivel pueden estimar parámetros de "múltiples niveles" simultáneamente.
- Por ejemplo, podemos modelar:
 - Variación dentro del sujeto (σ).
 - Promedios para cada sujeto ($L_{[i]}$).
 - Variación entre sujetos (σ_L).



Representing Factors with Many Levels

- Cada nivel de un factor necesita un predictor.
- Matemáticamente, esto es una matriz de 1s y 0s.
- Tenemos (aproximadamente) 15 predictores para nuestros 15 niveles de oyente.

$$\mu_{[i]} = \text{Intercept} + L_{[1]} \cdot 0 + L_{[2]} \cdot 1 + L_{[3]} \cdot 0 + \dots + L_{[15]} \cdot 0$$

$$\mu_{[i]} = \text{Intercept} + L_{[2]}$$

Representing Factors with Many Levels

- Representaremos los parámetros que representan los niveles de un factor como vectores.
- Estos serán seleccionados por nuestros predictores del mismo nombre.
- Esto significa: "Nuestro valor previsto para la prueba i , $\mu_{[i]}$, es la suma del intercepto del modelo y el coeficiente L indexado por el valor del predictor L para el ensayo i ($L_{[i]}$)."

$$\mu_{[i]} = \text{Intercept} + L_{[L_{[i]}]}$$


$$L_{[1]} = 2, L_{[2]} = 4, L_{[3]} = 1, \dots$$

$$\mu_{[i=1]} = \text{Intercept} + L_{[L_{[i=1]}]} = \text{Intercept} + L_{[2]}$$

$$\mu_{[i=2]} = \text{Intercept} + L_{[L_{[i=2]}]} = \text{Intercept} + L_{[4]}$$

$$\mu_{[i=3]} = \text{Intercept} + L_{[L_{[i=3]}]} = \text{Intercept} + L_{[1]}$$

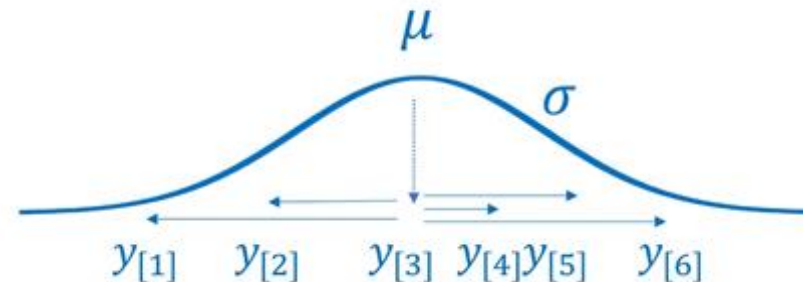
...

Strategies for Estimating Factors with Many Levels

- Sus factores pueden tener mas que 10 niveles (¡o más de 100!).
- Esto puede resultar en un gran número de parámetros.
- Hay tres maneras generales para tratar los factores con muchos niveles:
 - Agrupación completa.
 - Sin agrupación.
 - Agrupación parcial (adaptativa).

Complete Pooling

- Tiras todo en una gran pila y actúas como si no tuvieras clústeres de datos individuales.
- Pierdes información útil (e.g., σ_L).
- Gran problema: es probable que los datos no sean independientes y los IC no serán fiables.



$$\text{height}[i] \sim N(\mu[i], \sigma)$$

$$\mu[i] = \text{Intercept}$$

Priors:

$$\text{Intercept} \sim N(176, 15)$$

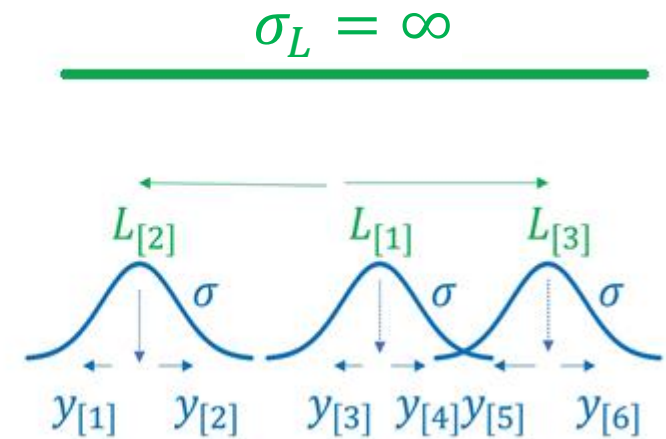
$$\sigma \sim N(0, 15)$$

No Pooling

- Reconoce los clústeres, pero no impone restricciones en sus valores.
- Pierdes información útil (e.g., σ_L).
- Problema mediano: Valores de parámetros entre clústeres (i.e. L) por lo general, no son totalmente independientes.

Level 2: Group Level

Level 1: Data Level



$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]}$$

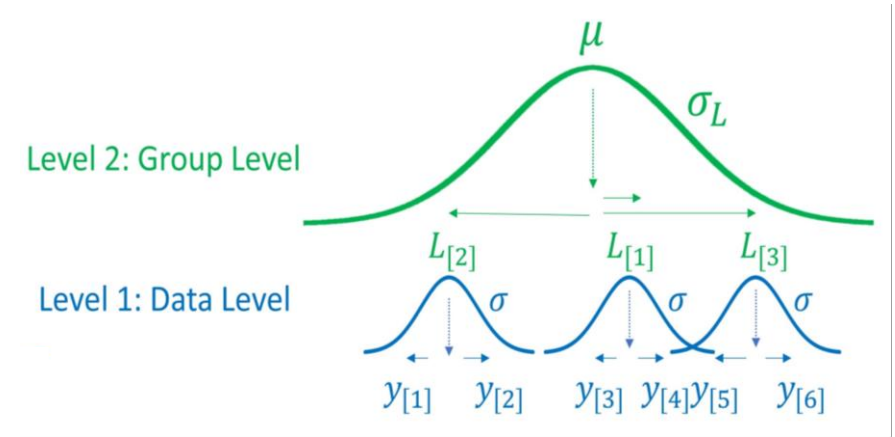
Priors:

$$L_{[\cdot]} \sim \text{uniform}(-\infty, \infty)$$

$$\text{Intercept} \sim N(176, 15)$$

Partial Pooling

- Reconoce los clústeres e impone restricciones probabilísticas a sus valores.
- Utiliza información en el "segundo" nivel (e.g, σ_L) para modelar.
- Gran beneficio: Más información y modelos que se comportan mejor.



$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]} \quad \leftarrow$$

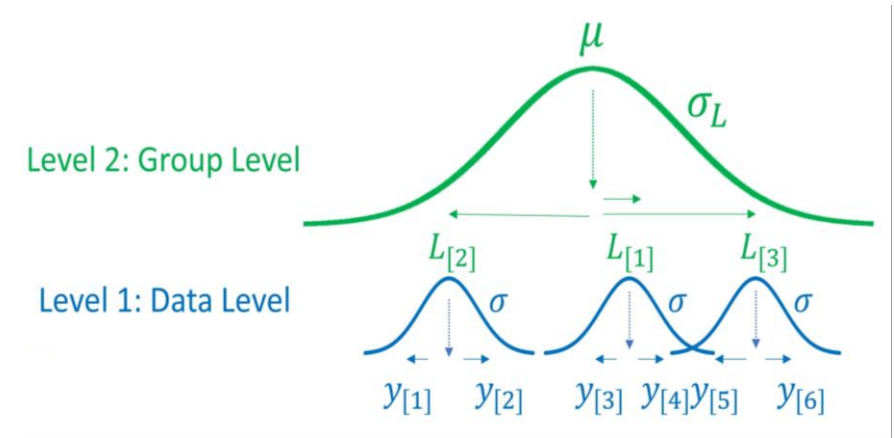
Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L) \quad \leftarrow$$

$$\text{Intercept} \sim N(176, 15)$$
$$\sigma \sim N(0, 15)$$
$$\sigma_L \sim N(0, 15) \quad \leftarrow$$

Adaptive Partial Pooling

- La agrupación parcial es cualquier modelo que utilice la misma información (e.g. σ_L) para diferentes parámetros.
- La agrupación parcial *adaptativa* es cuando un modelo estima estas restricciones (e.g. σ_L) de los datos.



$$\text{height}[i] \sim N(\mu[i], \sigma)$$
$$\mu[i] = \text{Intercept} + L[L[i]] \quad \leftarrow$$

Priors :

$$L[\cdot] \sim N(0, \sigma_L) \quad \leftarrow$$

$$\text{Intercept} \sim N(176, 15)$$
$$\sigma \sim N(0, 15)$$
$$\sigma_L \sim N(0, 15) \quad \leftarrow$$

Comparison of Approaches

Agrupación completa

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept}$$

Priors:

$$\text{Intercept} \sim \text{N}(176, 15)$$
$$\sigma \sim \text{N}(0, 15)$$

Sin agrupación

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]}$$

Priors:

$$L_{[\cdot]} \sim \text{uniform}(-\infty, \infty)$$

$$\text{Intercept} \sim \text{N}(176, 15)$$

Agrupación parcial

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]}$$

Priors :

$$L_{[\cdot]} \sim \text{N}(0, \sigma_L)$$

$$\text{Intercept} \sim \text{N}(176, 15)$$

$$\sigma \sim \text{N}(0, 15)$$

$$\sigma_L \sim \text{N}(0, 15)$$

Hyperpriors

- Los priores para tus priores.

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + L_{[i]}$$

Priors :

$$L_{[.]} \sim \text{N}(0, \sigma_L)$$



$$\text{Intercept} \sim \text{N}(176, 15)$$

$$\sigma \sim \text{N}(0, 15)$$

$$\sigma_L \sim \text{N}(0, 15)$$



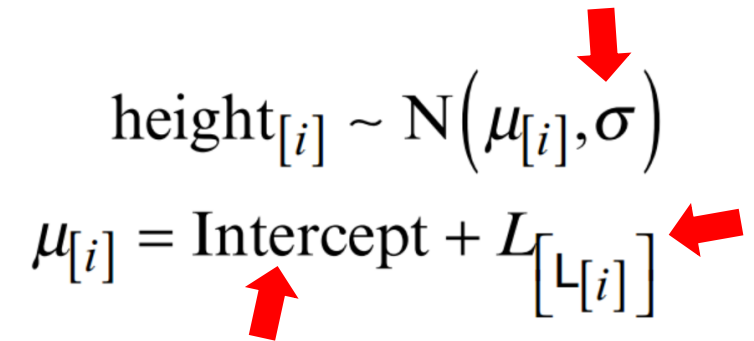
$$P(L_{[.]} | y) = \frac{P(y | L_{[.]}) \cdot P(L_{[.]})}{P(y)}$$

$$P(L_{[.]}, \sigma_L | y) = \frac{P(y | L_{[.]}, \sigma_L) \cdot P(L_{[.]}, \sigma_L)}{P(y)}$$

$$P(L_{[.]}, \sigma_L | y) = \frac{P(y | L_{[.]}) \cdot P(L_{[.] | \sigma_L) \cdot P(\sigma_L)}{P(y)}$$

What Gets a Prior?

- Todos los parámetros estimados necesitan priores.
- Por ejemplo:
 - L y σ_L necesitan priores porque están estimados.
 - La desviación estándar de σ_L es 15. No se estima.

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]}$$


Priors :

$$L_{[\cdot]} \sim \text{N}(0, \sigma_L)$$

$$\text{Intercept} \sim \text{N}(176, 15)$$

$$\sigma \sim \text{N}(0, 15)$$

$$\sigma_L \sim \text{N}(0, 15)$$

Data and Research Questions

```
# load book package and brms
library (bmmb)
library (brms)

# load and subset experimental data
data (exp_data)
men = exp_data[exp_data$C_v=='m',]
```

- **L**: An integer from 1 to 15 indicating which *listener* responded to the trial.
- **height**: A floating-point number representing the *height* (in centimeters) reported for the speaker on each trial.

```
head (men)
##           L C height R  S C_v  vt1  f0 dur G A G_v A_v
##  93      1 m  169.9 a 47    m 14.8 172 339 m a    m    a
##  95      1 m  173.5 a 48    m 15.6 108 236 m a    m    a
##  97      1 m  172.0 a 49    m 15.5  96 315 m a    m    a
```

Data and Research Questions

Vamos a responder de nuevo a las siguientes preguntas, esta vez con un modelo multinivel legítimo:

(1) ¿Qué tan alto suena el hombre adulto promedio?

(2) ¿Podemos establecer límites a las alturas aparentes medias creíbles en función de los datos que hemos recopilado?

The Model Formula

- Nuestro modelo anterior decía: "predecimos la altura usando un solo intercepto general".

```
height ~ 1
```

- No queremos modelar un intercepto, sino un intercepto para cada oyente.
- Queremos entender la distribución del intercepto dado el sujeto.

```
P(Intercept | Listener)
```

The Model Formula

- Para modelar interceptos dependientes del oyente con agrupación parcial adaptable, usamos esta fórmula:

```
height ~ 1 + ( 1 | L)
```

- Todo lo que pongas entre paréntesis se ajusta con agrupación parcial.

```
(Predictor|Grouping factor)
```


Description of the Model

$$\text{height}_{[i]} \sim \text{N}(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[\mathcal{L}_{[i]}]}$$

We expect height judgments to be normally distributed around the expected value for any given trial, $\mu_{[i]}$, with some unknown standard deviation σ . The expected value for a trial is equal to a fixed overall average (Intercept) and some value associated with the individual listener who made a perceptual judgment on the trial ($L_{[\mathcal{L}_{[i]}]}$). The listener coefficients ($L_{[\cdot]}$) were modeled as coming from a normal distribution with a mean of zero and a standard deviation (σ_L) that was estimated from the data.

Priors :

$$L_{[\cdot]} \sim \text{N}(0, \sigma_L)$$
$$\text{Intercept} \sim \text{N}(176, 15)$$
$$\sigma \sim \text{N}(0, 15)$$
$$\sigma_L \sim \text{N}(0, 15)$$

Decomposing Variation

Modelo 'Uninivel' $\longrightarrow \sigma_{\text{total}}^2 = \sigma^2$

Modelo multinivel $\longrightarrow \sigma_{\text{total}}^2 = \sigma_L^2 + \sigma^2$

Specifying Priors

- `Intercept`: This is a unique class, only for intercepts.
- `sd`: This is for standard deviation parameters related to ‘batches’ of parameters, e.g. `sd(Intercept)` for L (σ_L).
- `sigma`: The data-level error term.

```
bmb::get_prior (height ~ 1 + (1|L), data = men) [, -c(7:9)]
```

##	prior	class	coef	group	resp	dpar	source
##	<code>student_t(3, 174.5, 7.1)</code>	<code>Intercept</code>					<code>default</code>
##	<code>student_t(3, 0, 7.1)</code>	<code>sd</code>					<code>default</code>
##	<code>student_t(3, 0, 7.1)</code>	<code>sd</code>		<code>L</code>			<code>default</code>
##	<code>student_t(3, 0, 7.1)</code>	<code>sd Intercept</code>	<code>L</code>				<code>default</code>
##	<code>student_t(3, 0, 7.1)</code>	<code>sigma</code>					<code>default</code>

Fitting the Model

```
# Fit the model yourself
model_multilevel = brms::brm (
  height ~ 1 + (1|L), data = men, chains = 4, cores = 4,
  warmup = 1000, iter = 3500, thin = 2,
  prior = c(brms::set_prior("normal(176, 15)", class = "Intercept"),
     brms::set_prior("normal(0, 15)", class = "sd"),
    brms::set_prior("normal(0, 15)", class = "sigma")))

# Or download it from the GitHub page:
model_multilevel = bmmmb::get_model ('4_model_multilevel.RDS')
```

Interpreting the Model

```
# inspect model
bmb::short_summary (model_multilevel)
## Formula:  height ~ 1 + (1 | L)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)    3.78     0.87    2.47    5.84
##
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept    173.8      1.02   171.8   175.8
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma        7.03      0.19    6.67    7.41
```

$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]}$$

Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$\text{Intercept} \sim N(176, 15)$$

$$\sigma \sim N(0, 15)$$

$$\sigma_L \sim N(0, 15)$$

Interpreting the Model

```
# inspect model
bmb::short_summary (model_multil)
## Formula: height ~ 1 + (1 | L)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Err
## sd(Intercept)    3.78    0.
##
## Population-Level Effects:
##           Estimate Est.Error
## Intercept    173.8    1.02
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95%
## sigma        7.03    0.19
```

```
# find mean height for each listener
listener_means = aggregate (height ~ L, data = men, FUN = mean)

# find the within listener standard deviation
# This is the within-talker 'error'.
listener_sigmas = aggregate (height ~ L, data = men, FUN = sd)

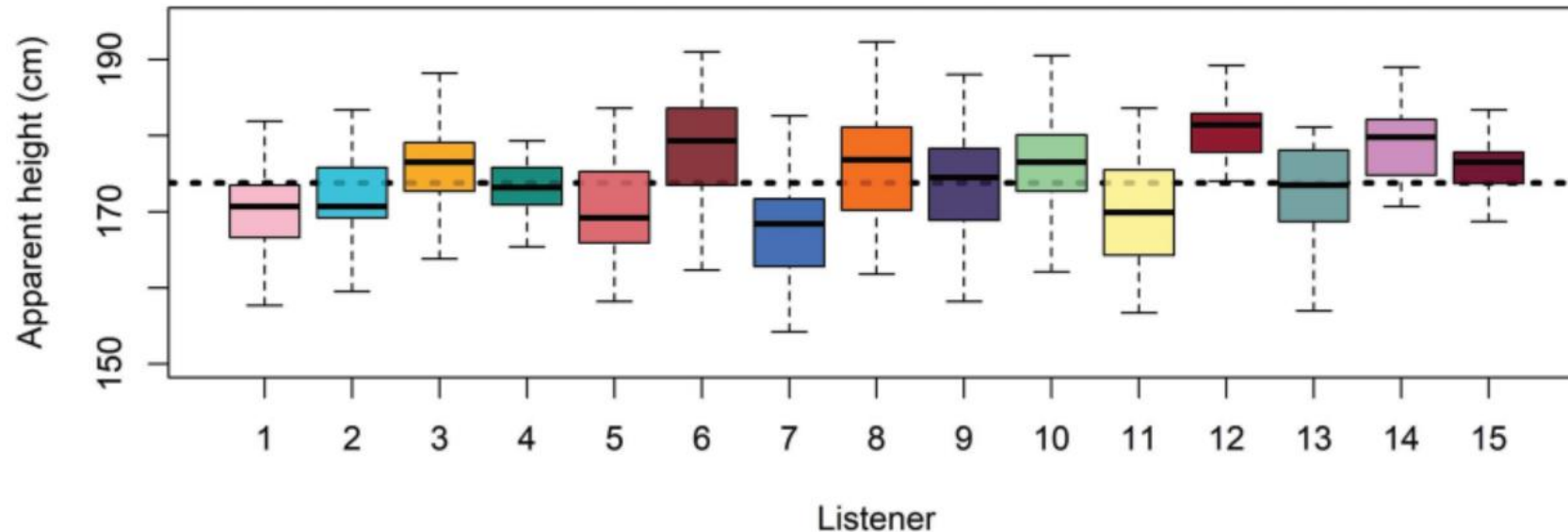
# the mean of the listener means corresponds to our Intercept
mean (listener_means$height)
## [1] 173.8

# the standard deviation of the listener means corresponds
# to 'sd(Intercept)', the estimate of the standard deviation
# of listener intercepts
sd (listener_means$height)
## [1] 3.594

# the average within-listener standard deviation corresponds
# to sigma, the estimated error
mean (listener_sigmas$height)
## [1] 6.822
```

Interpreting the Model

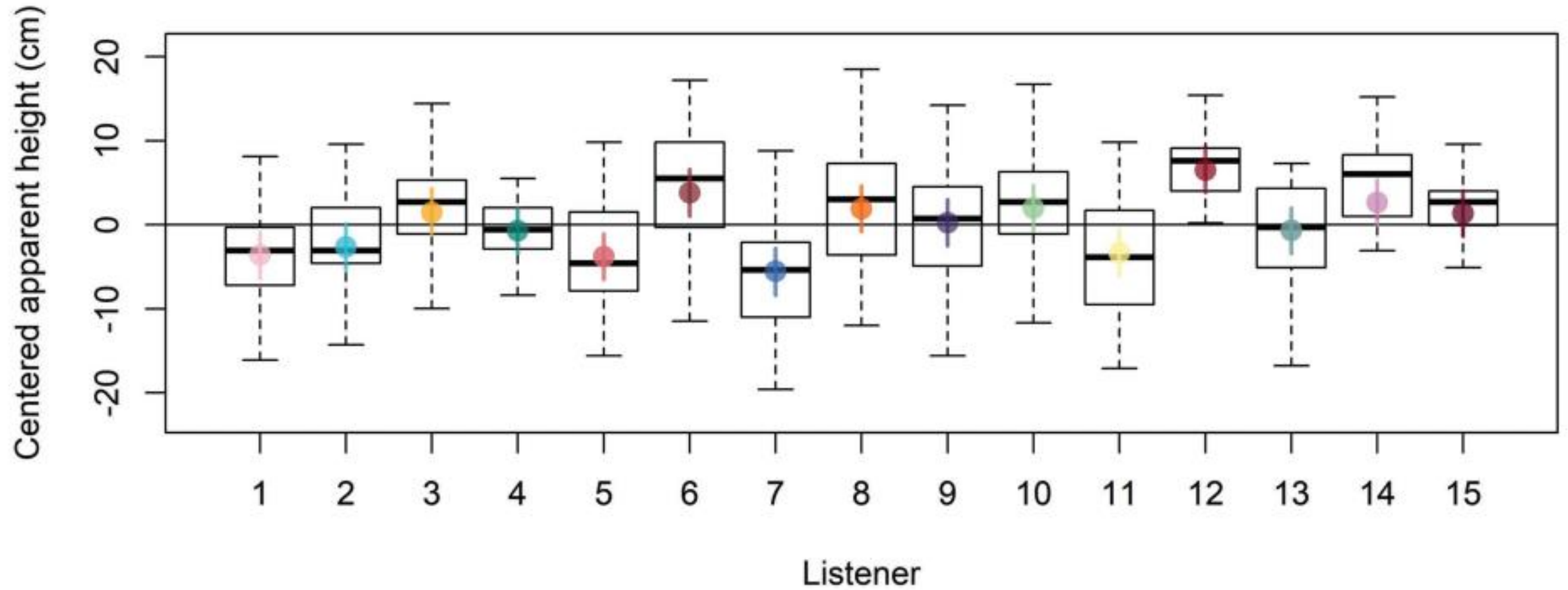
```
## Group-Level Effects:  
## ~L (Number of levels: 15)  
##           Estimate Est.Error 1-95% CI u-95% CI  
## sd(Intercept)      3.78      0.87    2.47    5.84  
##  
## Population-Level Effects:  
##           Estimate Est.Error 1-95% CI u-95% CI  
## Intercept      173.8      1.02   171.8   175.8  
##  
## Family Specific Parameters:  
##           Estimate Est.Error 1-95% CI u-95% CI  
## sigma          7.03      0.19    6.67    7.41
```



'Random' and 'Fixed' Effects

- Muchas definiciones inconsistentes de estos términos.
- En la práctica:
 - Los efectos "fijos" se estiman utilizando ninguna agrupación, o la agrupación es mínima.
 - Los efectos aleatorios suelen ajustarse con la agrupación parcial adaptativa.

Inspecting the 'Random' Effects



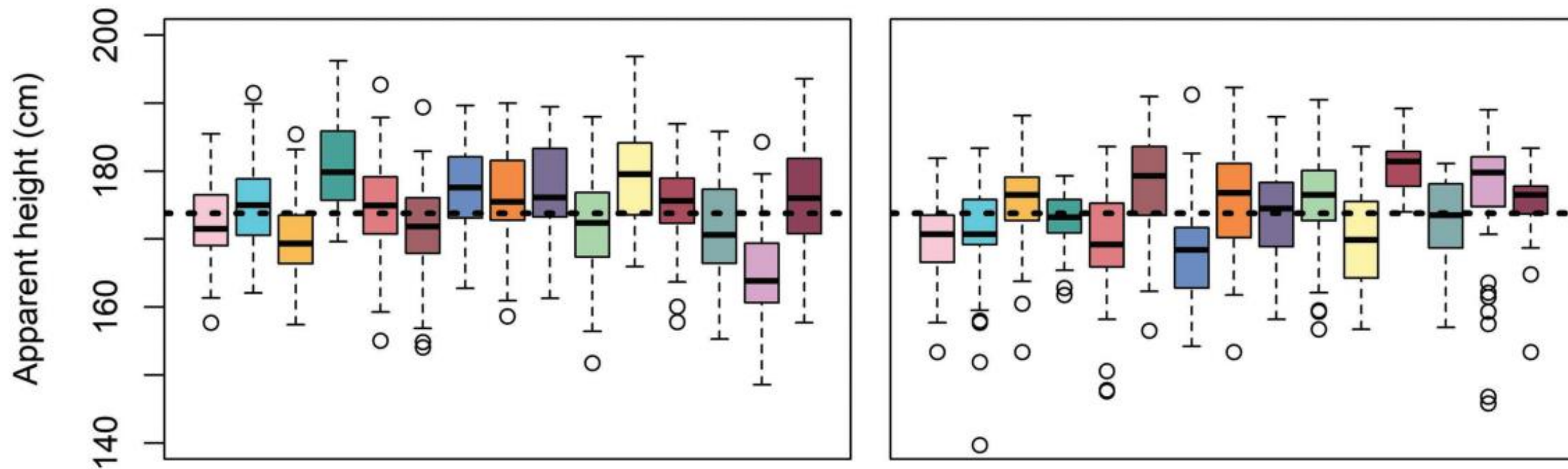
Simulating Data

```
# skip this line if you want a new simulated data set.  
set.seed(1)  
# this is the value of our intercept  
Intercept = 174  
# this is a vector of 15 listener effects  
L_ = rnorm (15, 0, 3.8 )  
# vector indicating which listener produced which utterance  
L = rep (1:15, each = 45)  
# this vector contains the error  
error = rnorm (45 * 15, 0, 7)
```

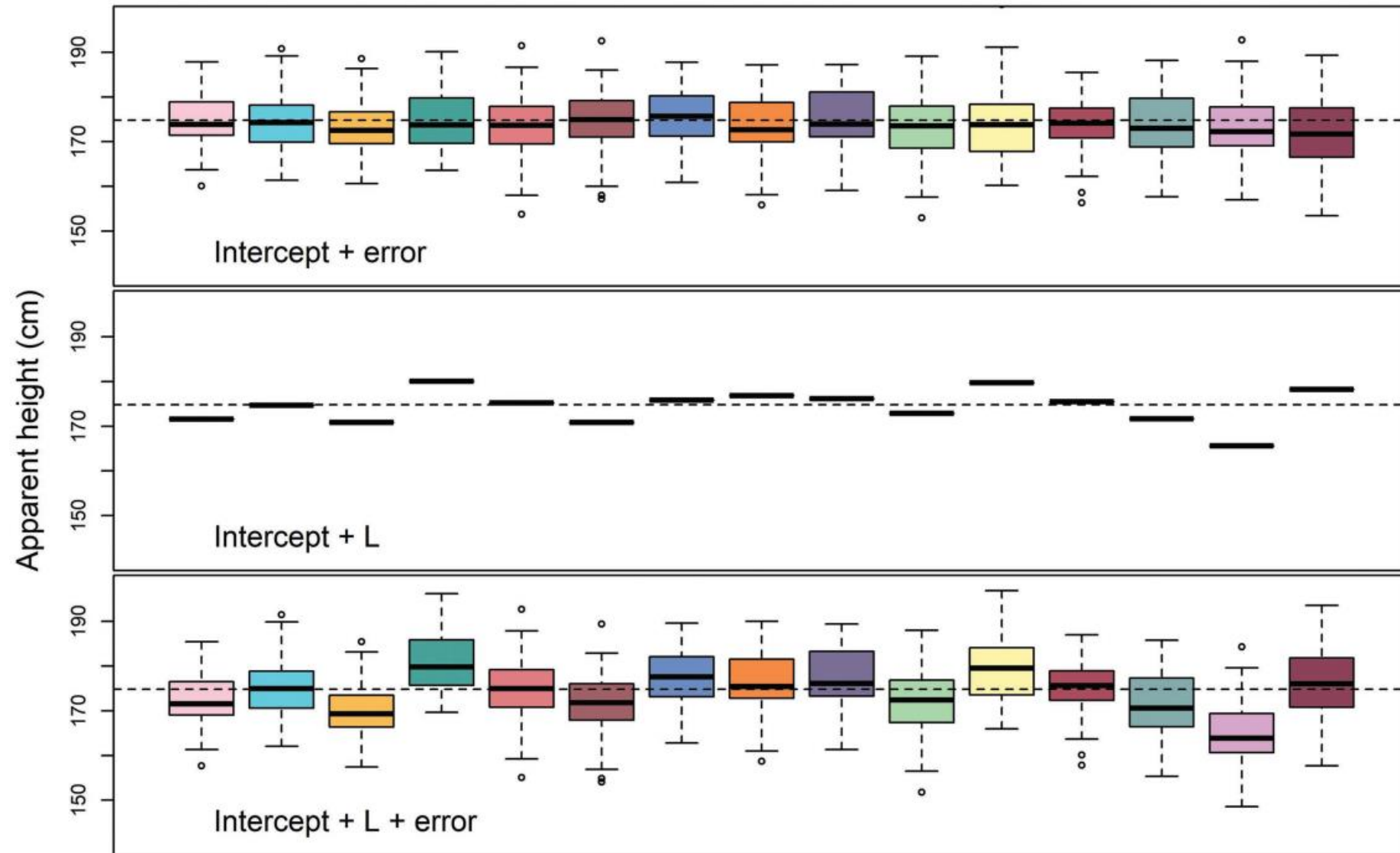
```
# the sum of an intercept, listener effects and random error  
height_rep = Intercept + L_[L] + error
```

```
# this fake data is missing between listener variation  
height_rep_1 = Intercept + error  
# this fake data is missing within listener variation  
height_rep_2 = Intercept + L_[L]
```

Simulating Data



Simulating Data



Simulating Data

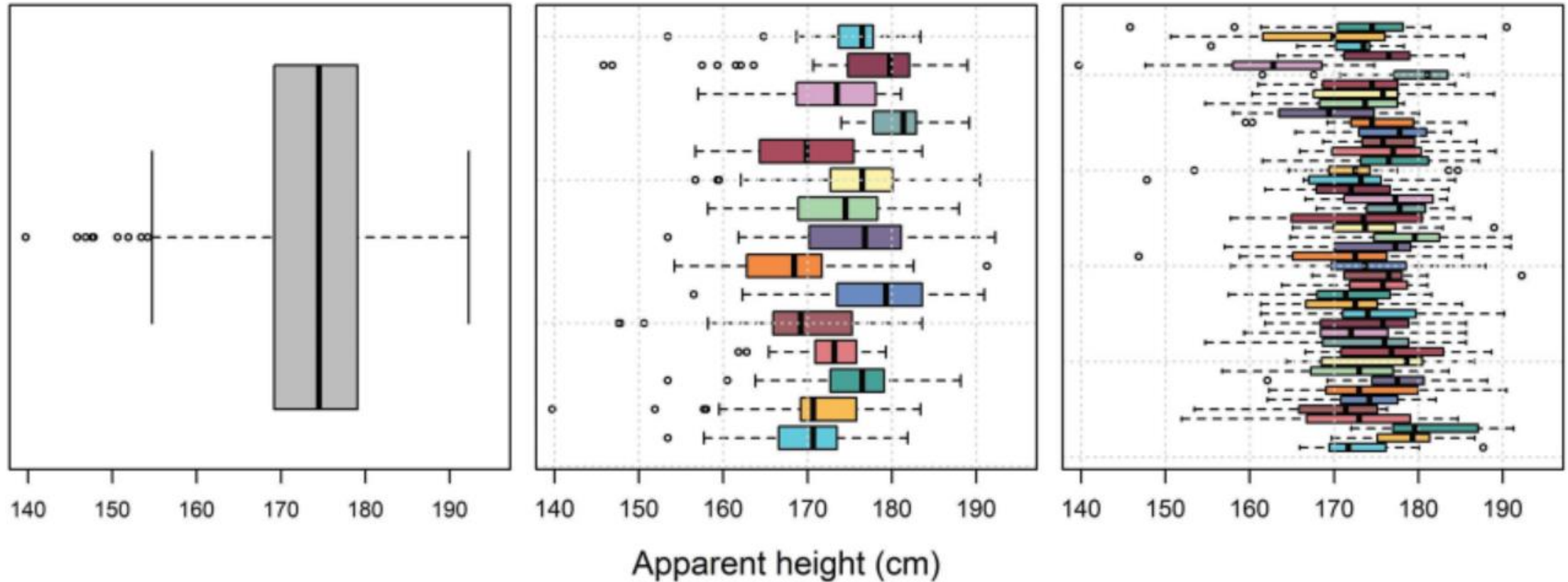
```
set.seed(1)
# do 10,000 replications
reps = 10000

# hold the replicated values of sigma_L
sigma_L_rep = rep(0, reps)
for ( i in 1:reps){
  Intercept = 173.8 # set the intercept
  L_L = rnorm (15, 0, 0) # zero between-listener variance
  L = rep (1:15, each = 45) # 45 responses from each of 15 listeners
  epsilon = rnorm (45 * 15, 0, 7.78) # generate random noise
  height_rep = Intercept + L_L[L] + epsilon # add up to components

# get replicated listener means
  L_rep_means = tapply(height_rep, L, mean)
  sigma_L_rep[i] = sd (L_rep_means) # find sigma of listener effects
}
```

```
quantile(sigma_L_rep)
##          0%          25%          50%          75%         100%
## 0.4429 0.9862 1.1337 1.2810 2.0813
```

Adding a Second Random Effect



Updating our Model Description

$$\text{height}_{[i]} \sim \mathcal{N}(\mu_{[i]}, \sigma)$$

$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]} + S_{[S[i]]} \leftarrow$$

Priors :

$$L_{[\cdot]} \sim \mathcal{N}(0, \sigma_L)$$

$$S_{[\cdot]} \sim \mathcal{N}(0, \sigma_S) \leftarrow$$

$$\text{Intercept} \sim \mathcal{N}(176, 15)$$

$$\sigma \sim \mathcal{N}(0, 15)$$

$$\sigma_L \sim \mathcal{N}(0, 15)$$

$$\sigma_S \sim \mathcal{N}(0, 15) \leftarrow$$

Fitting our Model

```
model_multilevel_L_S =  brms::brm (  
  height ~ 1 + (1|L) + (1|S), data = men, chains = 4, cores = 4,  
  warmup = 1000, iter = 3500, thin = 2,  
  prior = c(brms::set_prior("normal(176, 15)", class = "Intercept"),  
            brms::set_prior("normal(0, 15)", class = "sd"),  
            brms::set_prior("normal(0, 15)", class = "sigma")))
```


Interpreting the New Information

```
bmbb::short_summary (model_multilevel_L_S)
## Formula:  height ~ 1 + (1 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      3.81      0.86    2.51    5.87
##
## ~S (Number of levels: 45)
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)      2.83      0.42     2.1    3.72
##
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept      173.8      1.12   171.6   176.1
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma          6.47      0.19     6.11    6.85
```



$$\text{height}_{[i]} \sim N(\mu_{[i]}, \sigma)$$
$$\mu_{[i]} = \text{Intercept} + L_{[L[i]]} + S_{[S[i]]}$$

Priors :

$$L_{[\cdot]} \sim N(0, \sigma_L)$$

$$S_{[\cdot]} \sim N(0, \sigma_S)$$


$$\text{Intercept} \sim N(176, 15)$$

$$\sigma \sim N(0, 15)$$

$$\sigma_L \sim N(0, 15)$$

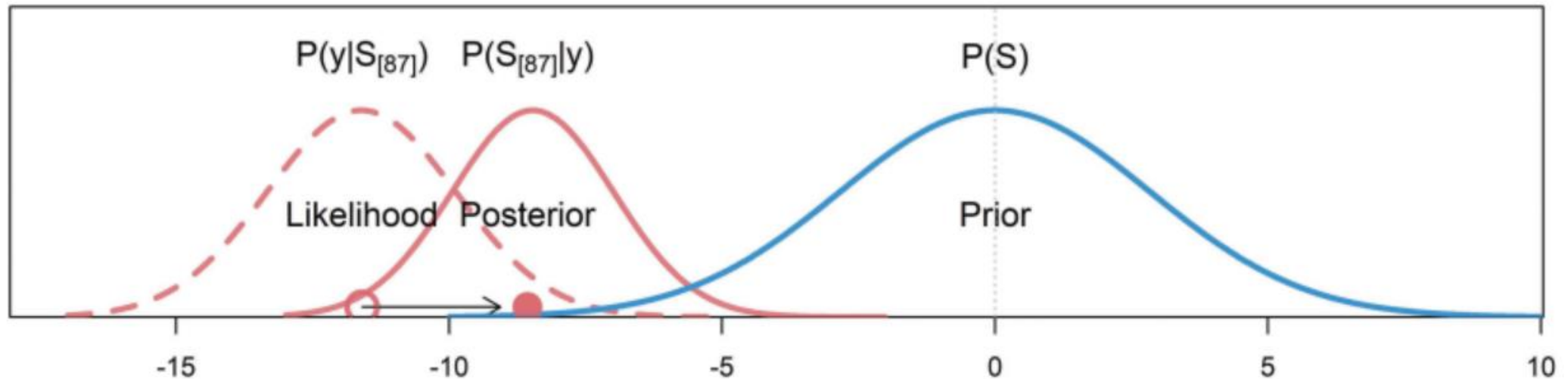
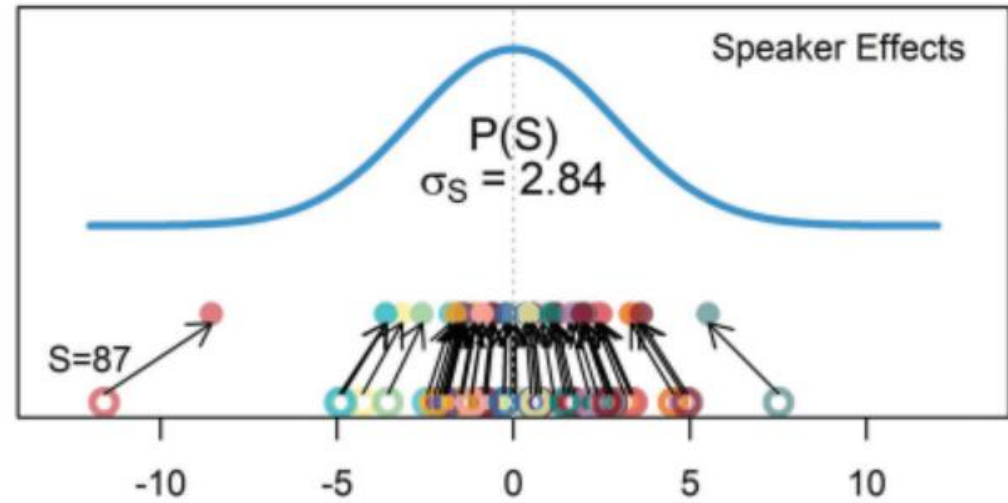
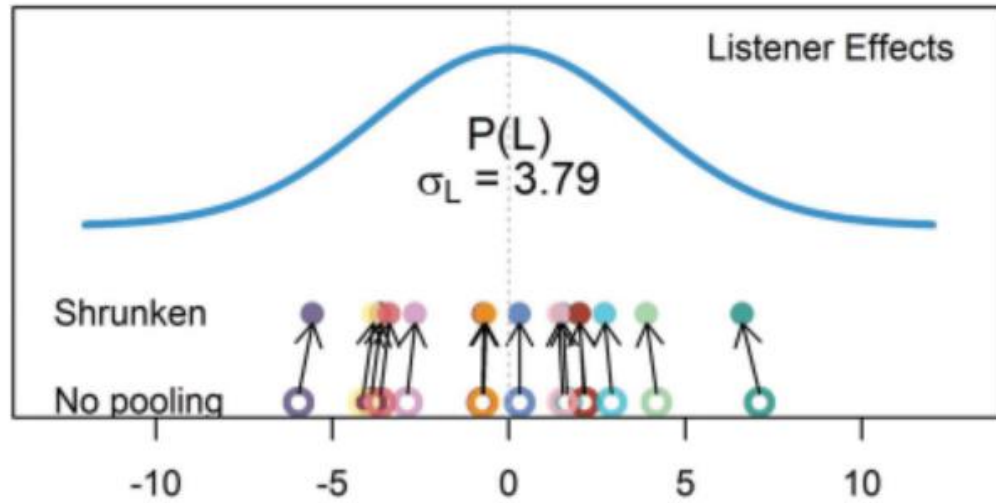
$$\sigma_S \sim N(0, 15)$$


Comparing Models

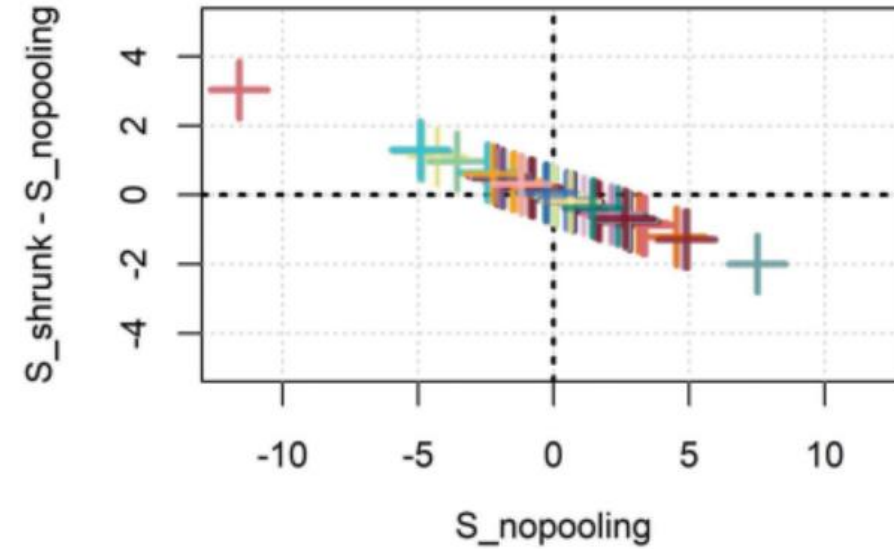
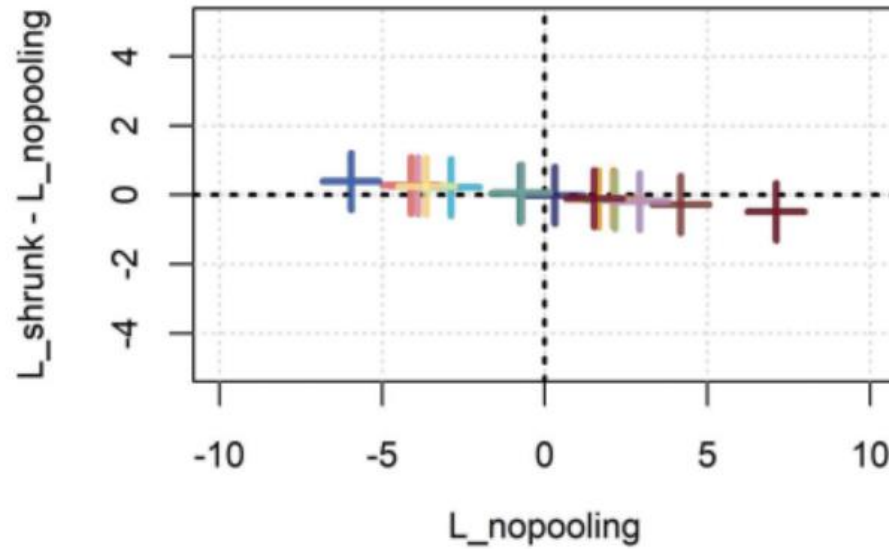
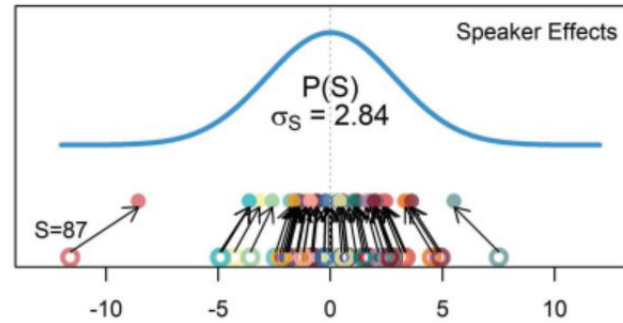
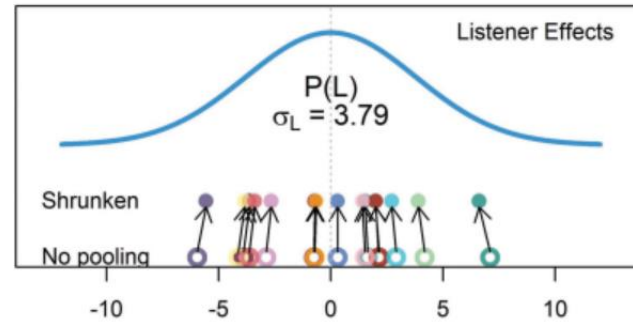
```
bmbb::short_summary (model_priors)
## Formula: height ~ 1
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept    173.8      0.31   173.2   174.4
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma         7.77      0.21    7.37    8.19
```

```
bmbb::short_summary (model_multilevel_L_S)
## Formula: height ~ 1 + (1 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)     3.81      0.86    2.51    5.87
##
## ~S (Number of levels: 45)
##           Estimate Est.Error 1-95% CI u-95% CI
## sd(Intercept)     2.83      0.42    2.1    3.72
##
## Population-Level Effects:
##           Estimate Est.Error 1-95% CI u-95% CI
## Intercept    173.8      1.12   171.6   176.1
##
## Family Specific Parameters:
##           Estimate Est.Error 1-95% CI u-95% CI
## sigma         6.47      0.19    6.11    6.85
```

Investigating 'Shrinkage'



Investigating 'Shrinkage'



Answering our Research Questions

```
bmb::short_summary (model_multilevel_L_S)
## Formula: height ~ 1 + (1 | L) + (1 | S)
##
## Group-Level Effects:
## ~L (Number of levels: 15)
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)    3.81    0.86    2.51    5.87
##
## ~S (Number of levels: 45)
##           Estimate Est.Error l-95% CI u-95% CI
## sd(Intercept)    2.83    0.42    2.1    3.72
##
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI
## Intercept    173.8    1.12    171.6    176.1
##
## Family Specific Parameters:
##           Estimate Est.Error l-95% CI u-95% CI
## sigma        6.47    0.19    6.11    6.85
```

Based on our model the average apparent height of adult males is likely to be 173.8 cm (s.d. = 1.1, 95% CI = [171.6, 176.1]). The estimated magnitude of the random error was 6.5 cm (s.d. = 0.2, 95% CI = [6.1, 6.9]). Systematic between-listener variation averages about 3.8 cm (s.d. = 0.9, 95% CI = [2.5, 5.8]), while systematic between-speaker variation averages about 2.8 cm (s.d. = 0.4, 95% CI = [2.1, 3.8]).

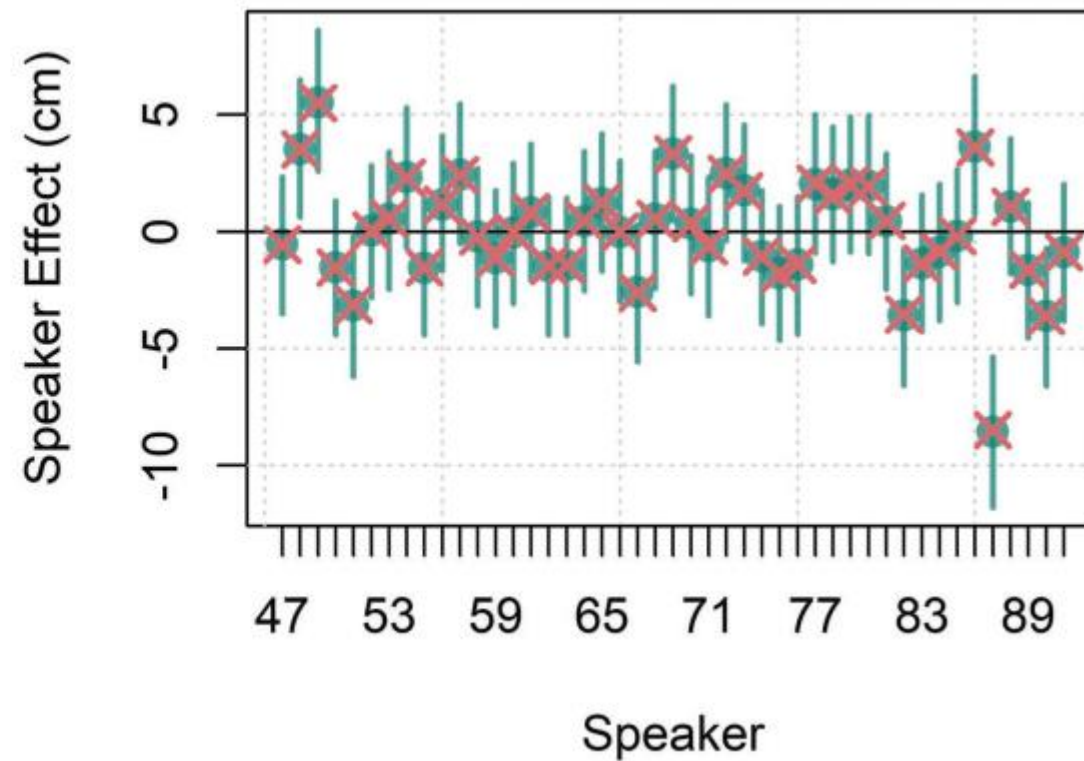
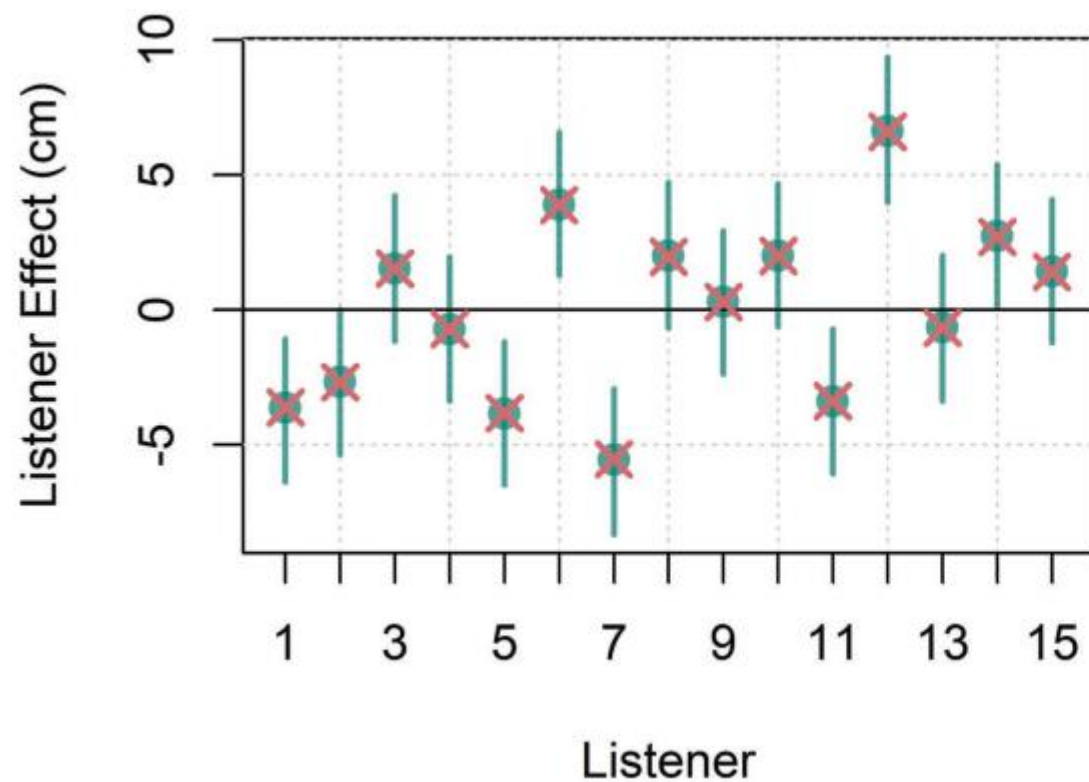
Traditionalists Corner: lmer

```
lmer_model = lme4::lmer (height ~ 1 + (1|L) + (1|S), data = men)
```

```
bmb::short_summary(model_multilevel_L_S)
## (1) Formula: height ~ 1 + (1 | L) + (1 | S)
##
## (2) Group-Level Effects:
##      ~L (Number of levels: 15)
##              Estimate Est.Error l-95% CI u-95% CI
## (3) sd(Intercept)      3.81      0.86      2.51      5.87
##
##      ~S (Number of levels: 45)
##              Estimate Est.Error l-95% CI u-95% CI
## (4) sd(Intercept)      2.83      0.42      2.1      3.72
##
##      Population-Level Effects:
##              Estimate Est.Error l-95% CI u-95% CI
## (5) Intercept      173.8      1.12     171.6     176.05
##
##      Family Specific Parameters:
##              Estimate Est.Error l-95% CI u-95% CI
## (6) sigma      6.47      0.19      6.11      6.85
```

```
summary (lmer_model)
##      Linear mixed model fit by REML ['lmerMod']
## (1) Formula: height ~ 1 + (1 | L) + (1 | S)
##      Data: men
##
##      REML criterion at convergence: 4527.4
##
##      Scaled residuals:
##              Min          1Q      Median          3Q          Max
##      -4.6205 -0.4868  0.0722  0.5700  2.7179
##
## (2) Random effects:
##      Groups      Name      Variance Std.Dev.
## (3) S      (Intercept)  7.593    2.756
## (4) L      (Intercept) 11.990    3.463
## (6) Residual              41.630    6.452
##      Number of obs: 675, groups: S, 45; L, 15
##
##      Fixed effects:
##              Estimate Std. Error t value
## (5) (Intercept)  173.788      1.015    171.3
```

Traditionalists Corner: Imer



Exercises

Use the data in 'exp_ex' to do one of the following. You may also use your own data to answer a related question. In any case, describe the model, present and explain the results, and include at least two figures.

1. **Easy**: Analyze the (pre-fit) model that's exactly like model_multilevel_L_S, except using the data in exp_ex (`bmmb::get_model("model_multilevel_L_S_ex.RDS")`).
2. **Medium**: Fit a model just like model_multilevel_L_S, but for the data from some other group, for either the original or big resonance levels.
3. **Hard**: Fit two models like model_multilevel_L_S for two groups, or for one group across resonance levels, and compare results across models.